Travel Planning Conversational Agent

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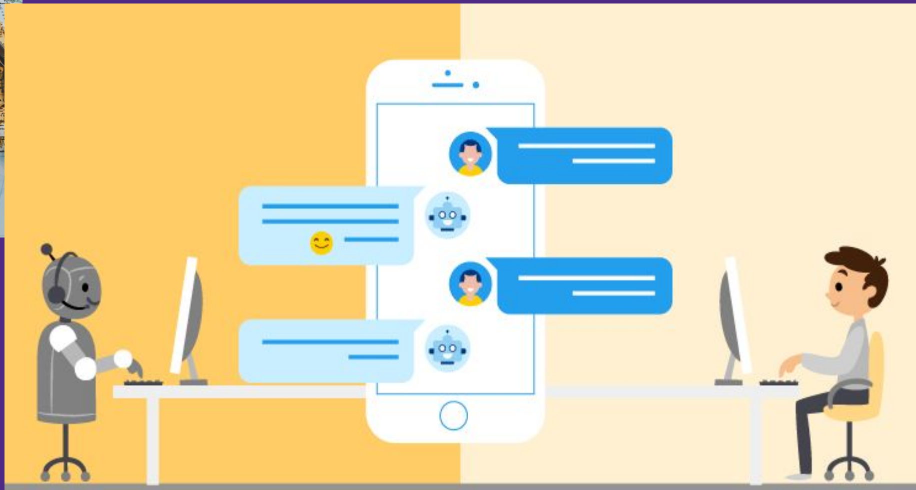
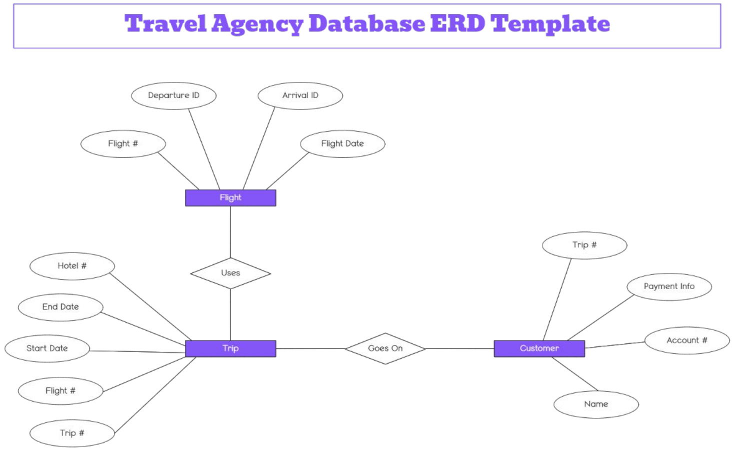
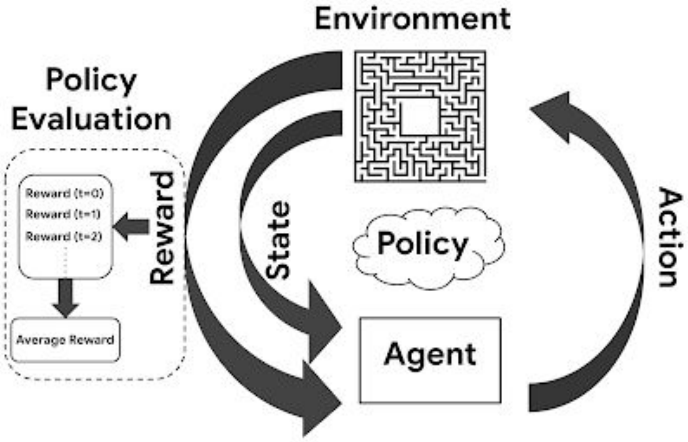


Project Overview

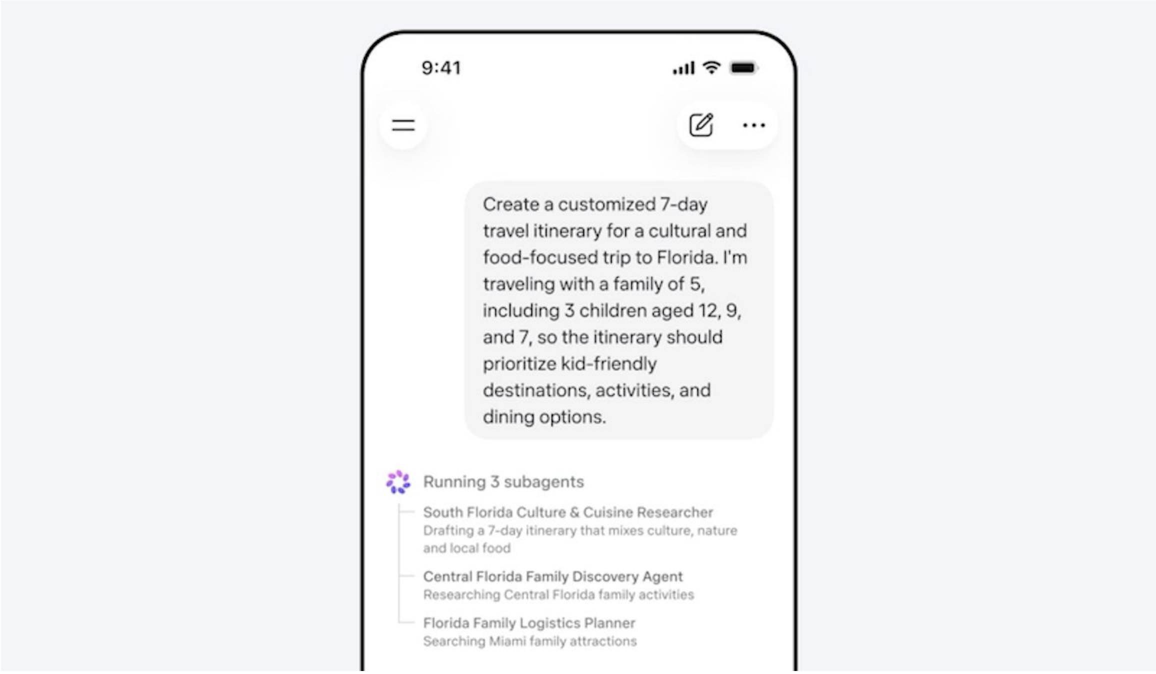
- Collects travel constraints through conversation
- Retrieves flights, hotels, restaurants, and activities
- Builds itineraries and simulates bookings
- Maintains dialogue context across multiple turns





Problem

- Travel planning is fragmented across flights, hotels, restaurants, and activities
- Users repeatedly enter preferences across multiple platforms
- Building a complete itinerary requires significant manual effort



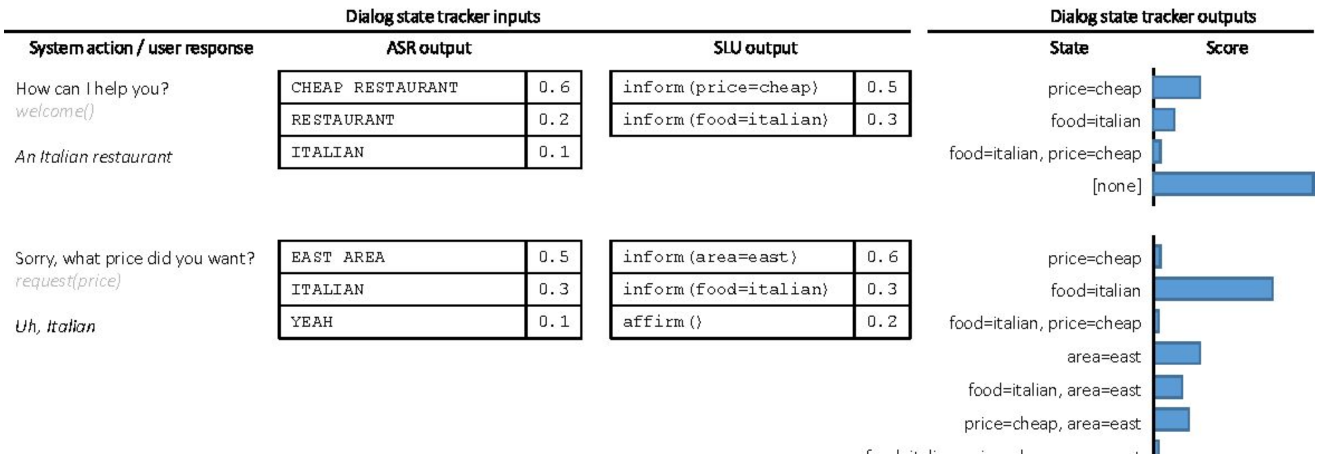
System Architecture

User Input → Dialogue State Tracker → Policy Manager → Travel Retrieval → Itinerary Planner → Response Generator

Pipeline implemented in agent.py orchestrating all system modules.

Dialogue State Tracking

- Hybrid DST implementation
- Regex extraction for dates, cities, travelers, and budget
- TF-IDF + Logistic Regression models trained on MultiWOZ 2.2
- Maintains structured state across conversation turns



Travel Database

Flights
3300

Hotels
120

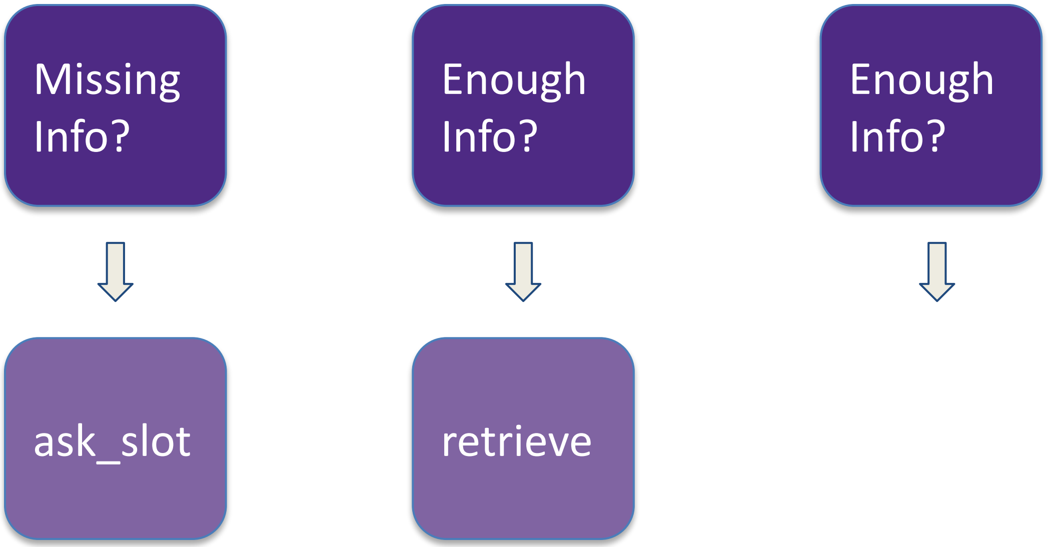
Restaurants
240

Activities
180



Policy Module

- Next-action prediction selects system behavior
- Actions include: ask_slot, retrieve, confirm, book, done
- Deterministic guardrails ensure valid conversation flow



End-to-End Example

User: I want to fly from New York to Miami...

System: Retrieves flights and hotels

User: Selects JetBlue + South Beach Grand

System: Confirms and books itinerary

Evaluation Framework

- Slot Accuracy
- Joint Goal Accuracy
- Retrieval Recall@K
- Policy Accuracy
- Task Completion Rate
- Average Turns to Completion

Results & Discussion

- [Insert evaluation graphs here]
- [Insert task completion metrics here]
- [Insert error analysis here]

Strengths & Limitations

- Strengths: modular architecture, grounded retrieval, interpretable pipeline
- Limitations: simulated bookings, limited database size, ambiguity handling

Future Work

- Real booking APIs
- LLM-enhanced response generation
- Personalized recommendations
- Larger travel databases